

# When Is a Risk Certificate Worth Its Sampling Cost?

A synthetic deepening of the recursive CVaR branch

Pathfinder supervisor-led research follow-up

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## Abstract

We instantiate the recursive paper’s conditional-risk obedience certificate in a two-signal routing model with a paid simulator, a reward budget and a bounded fallback. This gives a conditional expected-cost comparison absent from the parent paper’s application-free statement. A fixed simulation grid shows useful deployment on a separated instance, persistent abstention near the feasibility boundary, and a cost penalty for oversampling. Knowing the safe alternative exactly reduces both the uncertainty and the access bill. This is a synthetic worked application, not field validation, a new QP generation, or an independently reviewed manuscript.

## 1 Research question and lineage

The parent is *Priced Certificates for Conditional-CVaR Obedience*, recursive Q1P2, clarified version 2 of 28 September 2026. It combines an inherited deterministic reward graph with a simultaneous realised-data confidence box. Its open questions include pricing counterfactual access and specifying an abstention fallback. This follow-up supplies those ingredients in a deliberately small example. It does not add evidence to the matched repeat-versus-recurse comparison, whose allocations are already complete.

The protocol was written before executing the simulations. Code and output are in `experiments/2026-09-28-risk-deepening/`. The result file binds the parent, protocol and script by SHA-256. Historical papers are unchanged. All claims below are either direct algebra in this declared model or an application of the parent’s stated confidence guarantee. No publication priority or exhaustive external literature claim is made.

## 2 A declared routing model

Signals  $s = 0, 1$  are equally likely. The fixed policy prescribes action  $A$  at signal 0 and  $B$  at signal 1. At signal  $s$ , the prescribed route has loss  $2X_s$ , with  $X_s \sim \text{Bernoulli}(p_s)$ . The alternative route has constant loss 1. Thus each action is prescribed somewhere, as required by the parent. The agent evaluates conditional loss using upper-tail CVaR with tail mass  $\beta = 1/2$ , and subtracts its state-independent action reward. Recommendation-following breaks utility ties. This is weak obedience, not unique or strict implementation.

Write  $c_s = \min\{4p_s, 2\}$  for the prescribed route’s true CVaR and  $d = r(A) - r(B)$ . Exact obedience is simply

$$c_0 - 1 \leq d \leq 1 - c_1.$$

It is feasible exactly when  $c_0 + c_1 \leq 2$ . Among nonnegative rewards, choose the point of this interval closest to zero. Its absolute value is the oracle cap  $R^*$ , and its expected payment is  $R^*/2$ .

The planner pays operating cost  $g = 0.2$  on deployment and expected action reward  $|d|/2$ . For evaluation we conservatively add a surcharge  $J = 2$  if any true obedience constraint fails, rather

than modelling the deviating agent's route. Reward cap is  $R = 0.6$ . Fallback costs  $F = 1$ , needs no agent obedience, and is always available. Consequently downstream evaluation cost is at most  $M = g + R/2 + J = 2.5$ . These costs are stipulated in a common unit; they are not derived from the agent's risk objective or measured operational losses.

Sampling uses a conditional simulator with price  $\kappa$  per loss draw. This assumes access to deviation outcomes. On-policy logs alone do not justify this simulator. Two access regimes are compared:

- Uniform: sample all four signal-action cells, charging  $4m$  draws. The parent's generic certificate treats every cell as uncertain, even though safe-cell observations in this example are constant.
- Known safe: the alternative's constant loss is known from the model or an enforceable service contract. Sample only the two risky cells, charging  $2m$  draws. This assumption is stronger than observing a constant value in a finite batch.

### 3 Certificate, fallback and an expected-cost bound

For  $N = 4$  or  $2$  respectively, the parent gives simultaneous radius

$$e = 4\sqrt{\frac{\log(2N/\delta)}{2m}}, \quad \delta = 0.05.$$

The empirical risky CVaR is exactly  $\hat{c}_s = \min(4K_s/m, 2)$ , where  $K_s$  is a binomial count. Put  $t = 2e$  in the uniform regime and  $t = e$  in the known-safe regime. Since sampled safe losses equal 1, the robust interval in this particular application is

$$\hat{c}_0 - 1 + t \leq d \leq 1 - \hat{c}_1 - t.$$

If this interval intersects  $[-R, R]$ , deploy using its point closest to zero. If  $\hat{c}_0 + \hat{c}_1 - 2t > 2$ , record a refutation and use fallback. Otherwise use fallback, distinguishing an uncertain budget from ordinary abstention. A failed robust budget check does not certify failure of the true budget.

On simultaneous coverage, deployment is obedient and a refutation is sound. This is not a guarantee on every batch. Suppose  $\gamma = (2 - c_0 - c_1)/2 > 0$ . Define

$$I = \begin{cases} 4e & \text{uniform,} \\ 2e & \text{known safe.} \end{cases}$$

If  $I < \gamma$  and  $R^* + I \leq R$ , every covered batch deploys and its cap is at most  $R^* + I$ . To see this, each robust difference constraint exceeds its true counterpart by at most  $I$ ; their sum is at most  $-2\gamma + 2I < 0$ . The two-action path potential increases by at most  $I$ , and the budget condition permits that potential. The uniform bound deliberately retains the parent's generic error-box argument; the deterministic safe observations could also permit sharpening.

Let  $q = 4m$  or  $2m$ . Since downstream cost on coverage is at most  $B = g + (R^* + I)/2 \leq 0.5 < M$ , splitting by the coverage event gives

$$\mathbb{E}[\text{total cost}] \leq \kappa q + (1 - \delta) \left( g + \frac{R^* + I}{2} \right) + \delta M. \quad (1)$$

This compares with the feasible prediction-free benchmark of always falling back at cost 1. It is not a ratio against an unknown optimal system. It also differs from the oracle-with-free-information benchmark, which can deploy at  $g + R^*/2$  whenever feasible and within budget.

For  $(p_0, p_1) = (0.3, 0.1)$ , we have  $R^* = 0.2$  and  $\gamma = 0.2$ . Known-safe access at  $m = 4000$  and  $\kappa = 10^{-5}$  satisfies the premises and gives (1) approximately  $0.579 < 1$ . Uniform access at  $m = 64000$  gives approximately 3.018, not a useful guarantee at that price; at  $\kappa = 10^{-6}$  the same uniform bound is  $0.714 < 1$ . These are sufficient bounds, not optimal access-complexity results.

## 4 Fixed-batch numerical experiment

The protocol fixes five probability pairs, six batch sizes from 250 through 256000, two access regimes and two sampling prices. Each scenario-size combination has 2000 independent batches, pairing regimes on the same risky observations. There are 60000 batch pairs and 120000 regime decisions. Binomial counts are sampled directly using Python 3.13.0 with seed 20260928. No normal approximation is used. Guarantees apply per fixed trial, not simultaneously across the grid; no adaptive stopping is claimed.

Table 1 shows the paid-reward instance at the higher sampling price. Mean costs include fallback and the stipulated error surcharge. The baseline without sampling costs exactly 1.

| $m$   | Access     | Deployment (%) | Mean cost | MC standard error |
|-------|------------|----------------|-----------|-------------------|
| 1000  | Known safe | 63.2           | 0.629     | 0.00676           |
| 4000  | Uniform    | 47.3           | 0.872     | 0.00681           |
| 4000  | Known safe | 100.0          | 0.428     | 0.00103           |
| 16000 | Uniform    | 100.0          | 0.990     | 0.00016           |
| 16000 | Known safe | 100.0          | 0.644     | 0.00100           |
| 64000 | Uniform    | 100.0          | 2.885     | 0.00008           |
| 64000 | Known safe | 100.0          | 1.593     | 0.00100           |

Table 1: Paid instance (0.3, 0.1), price  $10^{-5}$  per draw. Deployment frequency is not correctness frequency.

Across the entire grid, six deployed rewards violate a true obedience constraint. They all occur in the paid scenario with known-safe access: one each at  $m = 1000, 4000, 16000, 64000$ , and two at 256000. All occur outside the recorded confidence box. There are 106 coverage failures across the paired regime decisions, and no incorrect refutations. The maximum observed false-decision frequency in a grid cell is 0.1%; neither that frequency nor zero errors in other cells proves coverage. The surcharge is retained in every reported cost, not removed as an outlier.

The free-reward scenario (0.1, 0.1) readily deploys without payments. The near-boundary scenario (0.35, 0.149) and exact boundary (0.35, 0.15) abstain in every tested batch under both regimes. For the infeasible scenario (0.35, 0.2), known-safe access refutes in 61.05% of batches at  $m = 4000$  and all batches at 16000; uniform access refutes in 48% at 16000 and all batches at 64000. Refutation is informative but cannot repay sampling in this one-decision model: fallback is still required, and could have been chosen for free.

## 5 What this deepening adds

First, specifying a fallback and a failure-path cap closes the parent’s missing expected-cost calculation, for this model only. The guarantee can be economically useful: equation (1) is below fallback cost for a declared instance, batch and sampling price.

Second, the access model materially changes the result. Known-safe access both halves the charged draws and removes uncertainty from deviation loss. These effects are combined here, not causally separated. It would be misleading to describe the gain as a uniformly better algorithm on equal information assumptions.

Third, more evidence is not always better for the planner. Once deployment is already reliable, the sampling bill grows linearly while the reward premium decreases only as the inverse square root of batch size. The grid illustrates this tradeoff; selecting the best observed grid cell would be post-selection, not a validated adaptive policy.

Finally, a small margin changes the practical question. At (0.35, 0.149),  $\gamma = 0.002$ . The known-safe sufficient margin condition alone requires more than 35 million samples per risky cell; the uniform version requires more than 162 million per cell. At either declared price those sufficient

sample bills already exceed the fallback. This does not prove that every valid certificate needs so many samples: it diagnoses the conservatism and economic limitation of this particular sufficient rule. A useful next investigation is a budget-aware sequential certificate with valid time-uniform coverage, or amortising one certificate over explicitly stationary repeated decisions. Neither is implemented here.

## 6 Verification and limits

Tests cover exact CVaR examples, interval signs, budget fallback, confidence-box corners, access accounting and the sufficient deployment premises. The simulation checks every definitive outcome against the true array, including failures. These checks supplement the algebra above. Results depend on a correct conditional simulator, stationary losses, known safe loss in the stronger regime, and the chosen cost model. There is no measured simulator cost, multi-type mechanism, report incentive analysis, or field experiment. No external review was performed.

This is *deepening*: pursuing an intermediate paper’s application and limitations. It is not another recombination and should not be counted as a new success in the historical repeat-versus-recurse comparison.